

Istanbul VS Alphen aan den Rijn

Istanbul

- ✓ Explain when air pollution occurs and from which level it becomes harmful or deadly
- ✓ Explain air quality standards and legal regulations
- ✓ Analyze air pollution levels in Istanbul and different districts
- ✓ Explain differences between air quality standards in Türkiye and Europe

Global Scale

Air pollution is an increasingly serious problem affecting villages, cities, countries and the entire world. Air pollution occurs when harmful substances such as particulate matter (PM10 and PM2.5), ozone (O₃) and nitrogen dioxide (NO₂) disturb the natural balance of the atmosphere.

Sometimes air pollution appears in the form of smog, which means that there is a temporary increase in polluted particles in the air. Vulnerable groups such as elderly people, children, individuals with respiratory diseases, athletes and people who perform heavy physical work are especially sensitive to smog exposure. During severe smog events, everyone can experience health complaints.

According to the World Health Organization (WHO), air pollution is always harmful, but it becomes especially dangerous when concentrations exceed recommended guideline values. Long-term exposure can lead to chronic diseases, physical health problems and a shortened life expectancy.

National and Regional Scale

In the Netherlands, strict air quality regulations are applied as part of European Union legislation. These rules are continuously updated to align closely with WHO guidelines. The Netherlands also participates in national clean air agreements aimed at reducing emissions from traffic and industry.

Türkiye follows its own air quality regulations, as it is not a member of the European Union. These standards are generally less strict than EU and WHO guidelines. Traffic density, industrial activities and fossil fuel use contribute significantly to air pollution, particularly in large cities.

Local Scale

Istanbul faces serious air pollution challenges due to its high population density, heavy traffic and industrial activities. Road transport, maritime transport and residential heating significantly affect air quality.

Districts such as *Esenyurt, Bağcılar, Küçükçekmece, Kağıthane, Ümraniye* and *Kadıköy* frequently experience moderate to poor air quality. Areas near the *E-5 Highway* and the *TEM Motorway*, as well as industrial zones, show the highest concentrations of particulate matter.

Difference Between Istanbul and Alphen aan den Rijn

There is a clear difference between air pollution standards in the Netherlands, including Alphen aan den Rijn, and those applied in Türkiye, including Istanbul. The Netherlands follows strict European Union air quality regulations that are closely aligned with the guidelines of the World Health Organization.

Alphen aan den Rijn benefits from strong monitoring systems and consistent enforcement of environmental regulations, resulting in generally good air quality. Türkiye applies more flexible standards, and in Istanbul high population density, intense traffic and industrial activities lead to higher concentrations of air pollutants.

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Alphen aan den Rijn

- ☒ ~~Explaining when there is air pollution, from what point toxic, deadly~~
- ☒ ~~Explain what the norm is in NL, regulations what is allowed / not allowed~~
- ☒ ~~Then indicate what Alphen is in and see different streets in Alphen~~
- ☒ ~~Explain the difference in standards~~

Global Scale

Air pollution is probably not unknown to you, both villages, cities, countries and even the whole earth has to deal with it more and more. Air pollution is therefore a growing problem, there is air pollution when harmful substances such as particulate matter, ozone and nitrogen dioxide disturb the atmosphere. Sometimes this air pollution turns into smog, temporarily more polluted particles are in the air. People who are vulnerable to poor air quality will certainly also be sensitive to light smog. Athletes and people who perform heavy physical work are also vulnerable to exposure to smog. With severe smog, anyone can get complaints. Air pollution is always harmful but only becomes harmful as soon as the concentration of substances exceeds the advisory values of the WHO (World Health Organization). Smog is really deadly when the air is constantly polluted and people are exposed here for a long time. This can lead to many physical complaints and chronic diseases. The service life can also be shortened by smog. In the Netherlands, the lifespan is on average 8 months shorter.

National and regional scale

In the Netherlands, we have different rules for limiting particulate matter and other substances in the air. For example, the Netherlands is in an agreement with the rest of the European Union, which together have made rules on this. Furthermore, the Netherlands also has a Clean Air Agreement and follows the WHO advice as well as possible. However, these last rules are violated by the ports in Rotterdam, which means that Schiedam and Vlaardingen are above these standards.

Local scale

Alphen aan den Rijn is also a member of the Clean Air agreement, but there are concerns about whether these standards are met in some places. For example, several maps clearly show that there is moderate air quality in many places. This means that the air quality is acceptable, but can cause mild health complaints among vulnerable people. In general, the air quality in Alphen is just outside the standards with occasional outliers at industrial areas and busy roads, such as the N207, the Prins Bernhardlaan and the industrial area bordering the Hoorn.

Difference Istanbul and Alphen aan den Rijn there is a clear difference between the standards for air pollution in the Netherlands (and therefore also

Alphen aan den Rijn) and Turkey (including Istanbul). The Netherlands follows as previously mentioned

the ever stricter EU standards so that they are ever closer to the opinions of the

Who's coming. Turkey, on the other hand, is not a member of the EU and therefore has its own regulations and therefore also has other standards. These are generally less strict, as a result, large cities such as Istanbul are more polluted by higher concentrations of traffic and industrial areas. The rules are also less enforced in Turkey. Turkey is therefore somewhat more flexible in the rules than the Netherlands, but is therefore also more polluting.

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